



Module 3

Muscular System

This module discusses muscle physiology. As anatomy and physiology is a very broad, all-encompassing subject, it is sometimes difficult to know where to start, especially when discussing sport-specific training methodologies.

Without a basic understanding of muscle physiology, there is no foundation on which to build a solid training program. In order to coach athletes of any sport, the ability to apply sports science to form an appropriate training regimen is an absolute requirement. It is not enough to rely on one's past sport-specific experience to coach athletes. Experience is a great asset. However, it requires integration with a working knowledge of sports science in order to be a truly effective and resourceful coach.

❖❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- ☐ Skeletal muscle structure, function, and terminology
- ☐ Contractile properties of a muscle
- ☐ Muscle origin and insertion
- ☐ Muscle fiber types
- ☐ Muscle fiber recruitment
- ☐ Muscle hypertrophy
- ☐ Nerve innervation of muscles
- ☐ Types of connective tissue
- ☐ Role of fascia
- ☐ Primary muscles and their actions (as noted in body images)
- ☐ Inner and outer core musculature
- ☐ Kinetic linking/muscle imbalances
- ☐ "Warming up" muscles
- ☐ Delayed Onset Muscle Soreness (DOMS)/Muscle Burn

❖❖ MUSCLE PHYSIOLOGY

TERMINOLOGY

Concentric muscle contraction: shortening of a muscle

- Flexing the elbow concentrically contracts the bicep muscle.

Eccentric muscle contraction: lengthening of a muscle

- Extending the elbow eccentrically contracts the bicep muscle.

Isometric muscle contraction: A muscle contraction occurs, but no change in muscle length results.

- Holding a weight in your hand with your elbow at 90 degrees with no motion is an isometric contraction.

Isotonic muscle contraction: A muscle contraction occurs, and the muscle shortens. This shortening of the muscle can occur at varying rates of speed.

- A push-up (the upward motion) represents an isotonic chest muscle (pectoralis major) contraction.

Isokinetic muscle contraction: when a muscle contraction occurs and the muscle shortens. The shortening/lengthening of the muscle occurs at a set speed. The speed of the muscle shortening/lengthening is usually controlled by an outside source.

- Performing seated leg extensions on a machine that controls the speed (set speed) of the movement via a motorized arm is an isokinetic contraction.

Hypertrophy (hi-per-tro-fee): increase in muscle size.

- A bodybuilder's muscles are the result of hypertrophy.

Atrophy: decrease in muscle size.

- Individuals on bed rest for long periods of time often experience muscle atrophy.

Agonist: muscle or muscles that are primarily responsible for movement around a joint; also called a *prime mover*

- The quadriceps muscles are the agonist muscles during leg extension.

Antagonist: muscle or muscles that act in opposition to the agonist muscles

- The triceps are the antagonist to the biceps.

The three types of muscle tissue in the body are:

- 1 **Cardiac muscle:** located only in the heart and is solely responsible for contracting/relaxing the heart to pump blood to the lungs and the rest of the body
- 2 **Smooth muscle:** involuntary muscles that surround organs for the purpose of protection against outside forces
- 3 **Skeletal muscle:** any muscle that is neither cardiac nor smooth. These are the muscles that provide movement of the body and account for 35–45 percent of a human's body weight (5). There are approximately 640 skeletal muscles in the body, most of which are grouped in pairs.

There are two types of muscle contractions: *voluntary* and *involuntary*.

- 1 **Voluntary Muscle Contractions:** These are muscle contractions that are controlled by the central nervous system (CNS) and the origin of the contraction is a conscious thought by the brain. An example of this is a bicep curl.
- 2 **Involuntary Muscle Contraction:** This type of muscle contraction is controlled by the autonomic nervous system (ANS) and occurs without conscious thought. An example of this is the contraction of the heart.

'Practice Makes Perfect'

This is a phrase erroneously used in relation to learning a skill. While repeated practice can make a movement pattern permanent (e.g., muscle memory), unless it is done correctly, the form will not be perfect.

SKELETAL MUSCLE STRUCTURE

The structure of skeletal muscle can be thought of as a thick cable on a suspension bridge. If you examined the cable that suspends a bridge, you would find that the large cable is made up of many smaller cables that, in turn, are made up of even smaller cables. At the smallest level, each wire strand is woven to make up a single cable that might be less than a millimeter thick. However, after bundling these wires together to create a cable, bundling those cables together to create a larger cable, and repeating this process many times, the resulting cable is able to support the road as well as the traffic.

Likewise, skeletal muscle is not a singular “cable” but rather a series of smaller cables that make up the whole muscle.

Below is an image (figure 3.1) that denotes the layers that make up a muscle, starting at the muscle fiber layer (17).

Structure of a Skeletal Muscle

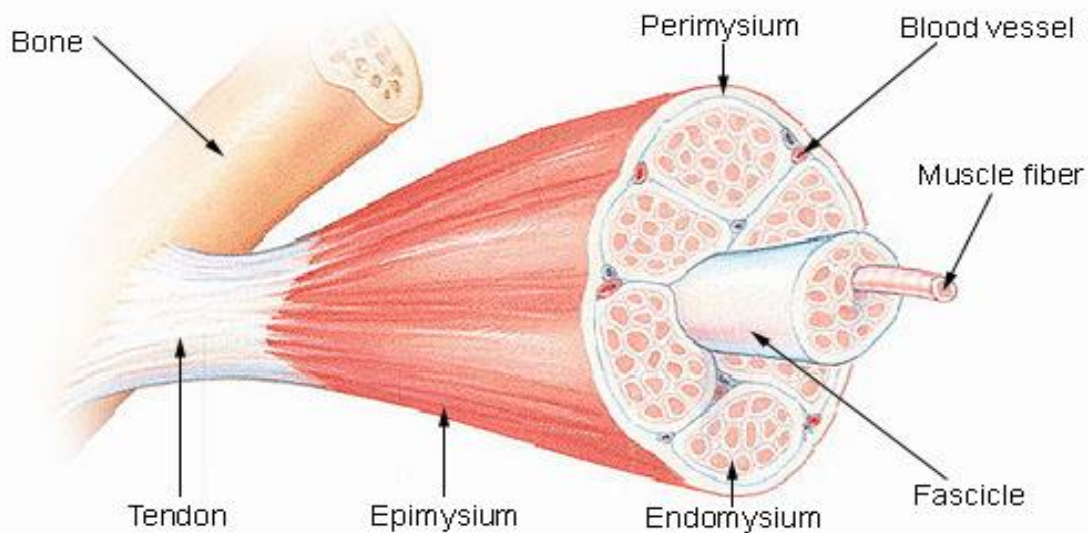


Figure 3.1 Skeletal Muscle Structure

http://training.seer.cancer.gov/module_anatomy/unit4_2_muscle_structure.html

Contractile Properties

To understand how a muscle contracts, it is helpful to break down the composition of a *muscle fiber* (i.e., muscle cell) via a hierarchy.

Muscle fiber is made up of *myofibrils*



Myofibrils are made up of *sarcomeres*



Sarcomeres are made up of *actin* and *myosin*

Myofibril: Contractile fibers that run along the length of a muscle fiber. The ability of a myofibril to contract is due to the sarcomere. When myofibrils contract, the muscle contracts.

Research in the realm of muscle fatigue suggests that the capacity of a myofibril to produce force may be affected by fatigue (556, 557).

Sarcomere: The sarcomere is a unit within the myofibril that is responsible for muscle contraction. In essence, it is the functional unit of the muscle fiber. A sarcomere contains three proteins types that have different roles. These roles and associated proteins are:

1. **Contractile:** *Actin* (thin filament) and *myosin* (thick filament) produce a contraction and shorten the sarcomere.
2. **Regulatory:** Made up of *tropomyosin* and *troponin* that allow or disallow the contractile process to take place.
3. **Structural:** Made up of *titin* and *myomesin* that keeps the contractile filaments (*actin* and *myosin*) aligned (18).

The shortening of the sarcomere, and therefore the muscle contraction, is caused by what is termed the **sliding filament model** (14). This model states that within the sarcomere, actin and myosin are alternately layered on top of one another. Myosin ratchets along the actin and causes the actin to shorten. This compression causes the sarcomere to shorten. Each sarcomere is connected to the next sarcomere by what is called a *Z line*, or *Z-disk*. When a muscle fiber contracts, all of the sarcomeres contract at the same time. This simultaneous contraction produces force at each end of the muscle fiber, causing a muscle contraction.

The speed and force of a muscle contraction relates to how much power is expended by a muscle. The greater the speed of a muscle contraction and/or the more muscle contractions there are, the greater the amount of ATP is used. This correlates to an increase in power being produced (705).

If the actin and myosin are too far apart or too close together, the contractile properties of the muscle will be greatly diminished. This is termed the *muscle length/tension relationship* and is described later in this module.

A sport-related example of this relationship would be the effect on the hamstrings when riding a bike with the saddle too low (hamstrings are too short to produce sufficient force) or too high (hamstrings are too long to produce sufficient force).

Myonuclei and Satellite Cells

The nucleus of a muscle fiber is called the myonuclei (481). There are multiple myonuclei per muscle fiber. It is theorized that the gain and loss of myonuclei within a muscle fiber is directly related to muscle fiber hypertrophy and atrophy, respectively (482). This directly relates to the theory that myonuclei are responsible for the regulation of protein synthesis of the surrounding area (482, 483).

Satellite cells, or myosatellite cells, are found in muscle. These cells typically lie dormant in the muscle until needed. The primary role of satellite cells is to assist in muscle hypertrophy, recovery from an injured muscle, or maintenance of muscle mass (484, 485). When activated, these cells proliferate and attach to existing muscle fibers or even fuse together to create new muscle fibers (486).

In simple terms, exercise that overloads a muscle (even stretching) causes muscle damage. As a result, satellite cells are activated to assist in muscle regeneration (481).

Muscle: Origin and Insertion

Muscles are anchored to bones in two places by *tendons*. These attachments to bone have two distinct connections. The *origin* is where the tendon anchors to a nonmoving bone. The other attachment point is called the *insertion*, and this is where the tendon attaches to the bone that moves.

Let us take, for example, the *pectoralis major* (figure 3.2). The origin is the *clavicle* (collarbone) and *sternum* (breastbone), and the insertion is the superior aspect of the *humerus* (top of the arm bone closest to the shoulder).

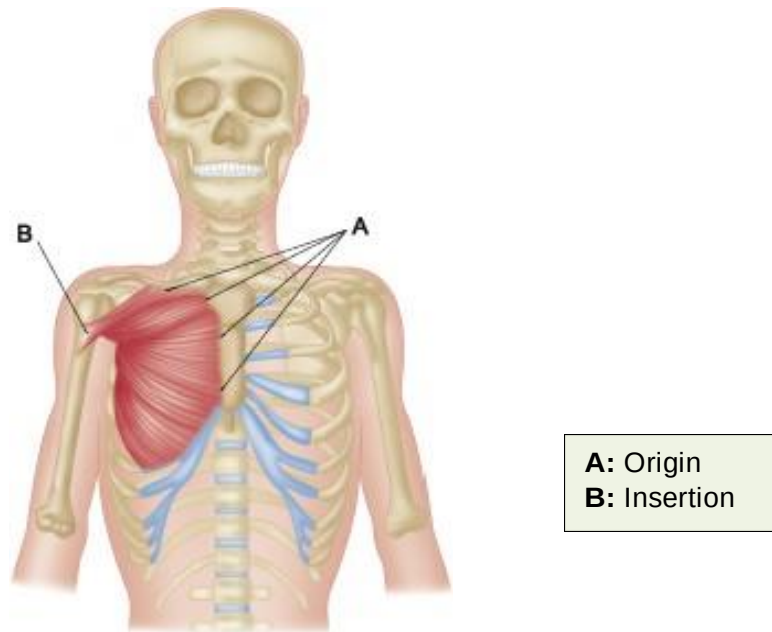


Figure 3.2 Skeletal Muscle Origin/Insertion

So what does this tell us? When looking at the direction of the muscle fibers and the location of the insertion and origin, it tells us that the muscle acts to adduct the humerus toward the midline of the body. Therefore, an exercise such as bringing the arms together in front of the chest (chest fly) work the pectoralis major in the range of motion specific to the direction of the muscle fibers and origin/insertion.

By understanding the insertion and origin of muscles, you will be able to understand what specific action the muscles perform and what exercises would be most effective for targeting a specific muscle or muscles.

Muscle Fiber Types

Not all muscle fibers are created equal, or it might be more accurate to say not all muscle fibers are created the same. One does not have to be overly observant to realize that the body type and, therefore, the muscle size of a competitive bodybuilder do not look like that of a distance runner. Why is this the case? Obviously, the training methodology of a bodybuilder versus that of a distance runner is quite different, but is that enough to make such a profound difference in

body type? There are many reasons that could play into the difference in physical appearance between these two athletes. Some of these reasons are:

- ☐ Diet
- ☐ Strength training program
- ☐ Cardiovascular training program

While the athletic performance and muscle size of an individual is certainly affected by the previously cited variables, absolute performance and muscle size of an individual is dictated largely by genetics.

There are two primary classifications of muscle fibers in the body, *fast twitch* (Type II-A and II-B) and *slow twitch* (Type I) (118).

- 1. Type I (Oxidative):** Type I fibers are characterized by their ability to fire repeatedly with minimal fatigue. They fire more slowly and with less force than Type II fibers and, as a result, are better geared toward endurance events, such as distance running.
- 2. Type II-A (Fast Oxidative Glycolytic):** Type II-A is an intermediate fast-twitch fiber. This fiber type is sort of a hybrid between Type I and Type II, meaning that its contractile properties are faster than Type I, but not quite as fast as Type II-B. Because of this, Type II-A fibers have midrange endurance capabilities. As this fiber type is between type I / II-B, it could also be termed, '*intermediate twitch*' due to it's contractile properties.
- 3. Type II-B (Fast Glycolytic):** Type II-B fibers are the classic fast-twitch fibers, which means their contractile properties are the fastest of all fiber types but they lack the ability to fire repeatedly without fatiguing. As a result, Type II-B fibers are best suited for short, explosive efforts.

The chart below demonstrates the characteristics of the muscle fiber types across a wide range of criteria (118).

Properties	Slow Twitch (Type I)	Fast Twitch (Type IIA)	Fast Twitch (Type IIB)
Size(Diameter)	Small	Medium	Large
ATP Synthesis (Energy Source)	Aerobic%(Triglycerides)	Aerobic/Anaerobic (CP/Glycogen)	Anaerobic (CP/Glycogen)
Mitochondria Density	High	Medium	Low
Capillary Density	High	Medium	Low
Endurance	High	Medium	Low
Force of Contraction	Low	High	High
Speed of Contraction	Low	High	High

%CP=Creatine|Phosphate

Figure 3.3 Muscle Fiber Classifications

While muscle size can be increased with strength training, the degree of hypertrophy is largely determined by the fiber type.

Some muscles within the body are composed of the same fiber type, regardless of genetics. This is largely dependent on the function of the muscle or muscles. For example, the *soleus* (calf muscle) is predominately made up of slow-twitch fibers (19). The reason for this is that the soleus is responsible for stability of the ankle and therefore it is constantly active and must have exceptional endurance properties.

Myostatin

While the type of muscle fiber plays a large role in determining the size of the fiber, a protein in the body called *myostatin* regulates how large a muscle fiber can become. This occurs by the myostatin regulating the amount of resources the muscles consume to grow in size (417). Individuals or animals with mutations to the myostatin protein have abnormally large muscles due to the lack of muscle size regulation (417).

Fast-twitch muscles produce more lactate than slow-twitch muscles. Therefore, the intensity of exercise and the fiber type ratio of an individual influence their lactate threshold (583).

Muscle Fiber Type Recruitment

In order to understand how muscles respond to the demands placed on them, it is important to know how muscle fibers are recruited – or engaged. Muscle fiber recruitment is the result of a neuromuscular reaction in which the central nervous system sends an electrochemical impulse to a particular muscle or muscles, at which point a contraction occurs. As noted previously, there are three types of muscle fiber types: Type I (oxidative), Type II-A (fast oxidative glycolytic), and Type II-B (fast glycolytic). When the intensity of an activity progresses from low or moderate to high, there is a hierarchy as to which types of muscle fibers are recruited. This is referred to as *Henneman's size principle* (356). First, Type I fibers are recruited, and as the intensity level rises, Type II-A and then Type II-B are recruited. If the intensity is increased over a very short period and to a very

high level (i.e., explosive-type movements), the central nervous system will recruit all three muscle types at the same time.

A muscle fiber cannot partially contract, or more specifically, a muscle fiber cannot contract with varying force depending on the effort required. A muscle fiber contracts either 100 percent or not at all. Therefore, the degree of muscle contraction is based on the number of muscle fibers recruited to perform an action, not the degree of contraction per muscle fiber.

Can Muscle Fibers Change Their Type?

Whether or not muscle fibers can change their type has long been debated, and to this day the debate has not been settled. In rare cases, muscles have demonstrated the ability to convert from Type I to Type II when the muscle was subjected to substantial deconditioning, in the case of injury, for example (21). There has not been much evidence to support the theory that muscles can change from Type I to Type II through changes in training routines (289), although the lack of verification might be due to insufficient research in this particular area.

While not attributable to a change in muscle fiber type, some research has shown that through specific types of training, the contraction speed of Type I fibers can increase (290, 291).

Regardless of a muscle fiber's ability to change type, people can always improve their performance even if it is at odds with their genetic muscle fiber type composition. For example, let us say your client has predominately fast-twitch muscle fibers and wants to compete in an endurance event such as a marathon. Through development and application of the proper training regimen, your client can become very proficient despite his or her genetic makeup to the extent that the individual can perform quite well in distance running events.

Muscle Hyperplasia vs. Muscle Hypertrophy

Hypertrophy refers to the increase in size of a muscle fiber whereas hyperplasia refers to the increase in the number of muscle fibers. While muscle hypertrophy has long been established as the method by which muscles increase in size, skeletal muscle hyperplasia is a much-debated topic. Skeletal muscle hyperplasia has been shown to exist in mammals as well as in human cardiac

muscle (415). However, evidence of skeletal muscle hyperplasia in humans is still largely inconclusive. A theorized reason for this is because the existence of skeletal muscle hyperplasia is impossible to detect via a standard muscle biopsy (416). Therefore, more research is required to determine with certainty if skeletal muscle hyperplasia exists in humans.

No Pain, No Gain?

It is widely accepted that muscles are damaged during intense exercise bouts and that they are rebuilt (i.e., remodeled, restructured) during the recovery phase, thus allowing for strength gains. However, detectable damage (i.e., plasma levels, pain, muscle soreness) to muscles is not a requirement for strength gains. In other words, just because an exercise does not cause pain does not mean that positive results aren't realized in regard to strength/performance.

A 2010 study by Flann et al. put two groups (trained/untrained) through training programs that elicited muscle damage in one group (untrained) but not in the other (trained) (683). It was found that both groups experienced similar increases in muscle rebuilding, strength gains, and muscle hypertrophy. Therefore muscle damage is not a requirement for muscle rebuilding and strength increases to occur.

Relationship Between Muscle Fiber Types to Fat Metabolism

Metabolism, or metabolic rate, is typically associated with how efficient one's metabolism is. While there are likely many variables that determine one's metabolism, several studies have demonstrated a link between muscle fiber type and fat metabolism (489, 490). Not surprisingly, the studies found that individuals with a high percentage of slow-twitch muscle fibers were more efficient at burning fat than those with high percentages of fast-twitch fibers.

As fat is the primary fuel for slow-twitch fibers, the greater the ratio of slow-twitch to fast-twitch fibers, the greater the capacity to burn fat is (489, 490). Therefore individuals with a high percentage of fast-twitch fibers often find losing fat to be more difficult as compared with someone with a high percentage of slow-twitch fibers.

SKELETAL MUSCLE FUNCTION AND CHARACTERISTICS

Muscle has four primary characteristics (16):

1. **Excitability:** responds to stimuli (e.g., nervous impulses)
2. **Contractibility:** shortens in length
3. **Extensibility:** stretches when pulled
4. **Elasticity:** returns to original shape and length after contraction

The three primary functions of muscle are:

1. **Heat Production** (up to 70 percent of body heat is generated via muscle tissue) (8)
2. **Maintain Posture**
3. **Movement**

As a running coach, your primary focus should be enhancing your client's ability to maintain posture and improve movement. Understanding that muscle tissue generates up to 70 percent of a human's body heat and how it relates to an athlete's performance and recovery is important to know, especially in regard to training and competing in cold and hot weather.

The contraction and relaxation of muscles has a profound effect regarding physiological changes in the body. When exercising at an intensity of 4 liters of oxygen/minute, a muscle's oxygen consumption increases approximately 70 times over that of the resting level (9)! In order to meet this demand, increased blood flow is channeled to the working muscles. When the working muscles contract, the blood flow to those muscles decreases, and when the muscles are relaxing, the blood flow increases. This pumping action helps to facilitate blood flow through the muscles and back to the heart. Capillaries, which are dormant under normal conditions, are utilized so that up to 4,000 of them deliver blood to each square millimeter of muscle (cross-section) during high-intensity exertion (10).

When a muscle is contracted to approximately 60 percent of its maximum contractile capacity, blood flow is occluded because of intramuscular pressure (10, 11).

Muscle Memory

Muscle memory is a term often used to denote an individual remembering how to perform a task, or the body's ability to regain muscular strength rapidly.

To be clear, muscles do not have a brain and thus do not have a memory. The ability of an individual to perform physical tasks by memory is the result of performing enough repetitions of a particular task that the motor skills required are memorized (277).

It is believed that learning a new skill (e.g., motor learning) occurs in the motor cortex of the frontal lobe of the brain. Once the skill has been memorized, the brain primarily utilizes the cerebellum to execute the skill (277).

Muscle memory is not a function of the muscle, but rather of the brain.

NERVE INNERVATION

Nerves are the pathways that deliver electrical impulses from the brain to the muscles, thereby eliciting a muscle contraction.

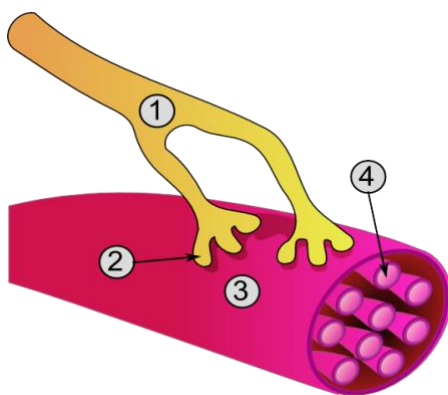


Figure 3.4 Nerve Innervation of Muscle Fiber

Neuromuscular junction (global view) # Axon # Synaptical junction # Muscle fiber # Myofibrils Drawn by fr:Utilisateur:Dake with Inkscape 0.42. {{GFDL}}

1. **Motor Neuron** (nerve)
2. **Synapse** (connection from nerve to muscle)
 - stimulate the muscle to contract
 - connection point of muscle fiber and synapse is the *neuromuscular junction*
3. **Muscle Fiber**
4. **Myofibril**

Of the three types of muscle tissues, only skeletal muscles have voluntary contractions. This means that the brain sends signals through the central nervous system (CNS) to the motor neurons that innervate muscle fibers and produce muscle contractions (6). This process is termed *neural signaling*. As individuals become better trained, their neural signaling can be improved (520).

A **motor unit** is a motor neuron and all the muscle fibers it innervates (figure 3.4).

Cardiac and smooth muscles rely upon the **autonomic nervous system (ANS)** to provide contractions. This means that cardiac and smooth muscle have involuntary contractions and therefore do not require the CNS to elicit a muscle contraction, and more to the point, the contractions occur without conscious thought. The ANS is part of the peripheral nervous system and has two primary areas of function:

1. **Sensory (afferent)**: an example of this would be physical pain experienced when you touch a hot surface
2. **Motor (efferent)**: an example of this are heart contractions

The ANS is broken up into the **sympathetic (SNS)** and **parasympathetic nervous systems (PNS)**. The actions of the SNS and PNS typically oppose each other. For example, the SNS increases cardiac output (the amount of blood pumped out of the heart in one minute), whereas the PNS decreases cardiac output (7). Therefore, the goal of the ANS is to keep the body in a state of equilibrium (*homeostasis*) by self-regulating itself via constant physiological adjustments.

As the nervous system is the catalyst for muscle contractions, an injury to a part of the nervous system may affect the ability of the related muscles to function properly. A common nerve issue is *sciatica*. Sciatica occurs when compression on the sciatic nerve causes pain and/or weakness in the leg to which the compressed sciatic nerve connects. This link demonstrates the relationship between the nerves and muscles.

CONNECTIVE TISSUE

Connective tissue is responsible for providing support and protection to the body and to support movement of the musculoskeletal system.

- **Tendons:** connect skeletal muscle to bone (e.g., the *Achilles tendon* connects the calf muscles to the heel – or *calcaneus*)
- **Ligaments:** connect bone to bone (e.g., the *patellar ligament* connects the knee – or *patella* – to the lower leg – or *tibia*)
- **Cartilage:** There are three types of cartilage: *hyaline (articular)*, *fibrocartilage*, and *elastic*. Relative to movement, the most important type to be aware of is hyaline cartilage. Hyaline cartilage is found on the surfaces of most joints and protects the joints while acting as a barrier between bones. Cartilage primarily comprises collagen and water.
 - **Labrum:** Found around the edge of the shoulder and hip joints. Its primary purpose is to keep the ball aspect of the humerus and femur within the shoulder and hip socket, respectively. It comprises *fibrocartilage*. Injury to the labrum is called a *labral tear*.
- **Fascia** (discussed below)

Shoulder

In addition to the rotator cuff muscles, the shoulder joint is stabilized by ligaments and the labrum. The head of the humerus sits in the *glenoid cavity*, which is also called the *glenoid fossa* (shoulder socket). The glenoid cavity is a relatively shallow cavity that allows the shoulder a large range of motion. The combination of the glenoid cavity and the head of the humerus is called the *glenohumeral joint*. Because of the shallowness of the glenoid cavity, the ligaments and labrum work to ensure that the shoulder does not dislocate or have excessive laxity (looseness).

Fascia

When most people learn about anatomy and how it applies to sports, the focus is typically on bones, muscles, and tendons/ligaments. This learning approach leaves out an important part of the whole picture – fascia. Fascia influences posture and facilitates body movement. If tension on the fascia is too high, it can cause pain and postural abnormalities as a result of restricted movement (270). Thomas Myers, author of *Anatomy Trains*, states, “Fascia is the missing element in the movement/stability equation” (330).

Fascia is a soft-tissue component of connective tissue (collagen fibers) that covers, connects, and separates all muscles, organs, bones, and nerves (331). Interestingly, while the fascia surrounds each of the aforementioned areas, it is actually one large, continuous tissue (330). Fascia is made up of multiple layers from superficial (just below the skin’s surface) to deep (covers internal organs). While thin, it is very strong and fibrous. If you have ever pulled the skin off raw chicken, you have encountered fascia. The fascia is the thin, near-white translucent membrane above the muscle and below the skin of the chicken. Perhaps the best way to think of fascia is a weblike structure that permeates the whole body.

Fascia provides support to the musculoskeletal system. According to the *International Fascia Research Congress (IFRC)*, while the basics of fascia are understood, the vast majority of information about fascia is largely not researched at all (187). The IFRC theorizes that one of the reasons for the lack of understanding of the fascia is that, because of its immense scope, researchers are often unable to divide it up into distinct areas on which to classify and focus. Additionally, one reason why the general population and even those who have taken anatomy classes disregard the fascia is that most anatomical diagrams/images show the body with the fascia removed (187).

Under normal conditions, fascia is pliable and flexible. However, when fascia is under stress because of trauma, dehydration, and biomechanical/neuromuscular issues, it becomes inflamed, which causes it to constrict and become less pliable and slippery. This leads to tightness, pain, and reduced range of motion. If the tightness is substantial enough, it can alter proper body mechanics. Fascia that is overly tight can produce tension up to 2,000 pounds per square inch (269)!

Like the body as a whole, fascia works better when it is hydrated. In individuals who are properly hydrated, the fascia is wet and slippery (330). This allows the

fascial surfaces to move unrestricted. However, when an individual is dehydrated, the fascia can become tight and “sticky,” thus reducing and altering normal movement patterns and ranges of motion.

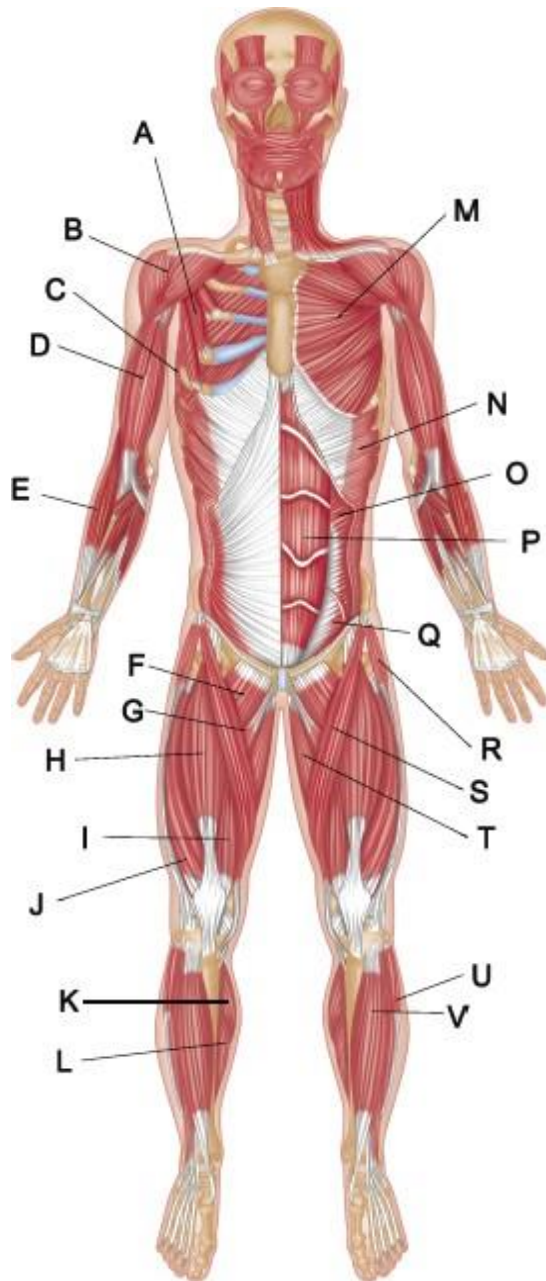
Diagnostic tests such as MRIs, CT scans, and X-rays do not show restricted fascia (332). As a result, individuals whose symptoms are caused primarily by restricted fascia often go undiagnosed or are incorrectly diagnosed (269). Fascia can contract with or independently of muscles (330).

The IFRC claims that fascia plays a major role in joint stability, biomechanical issues, chronic stress issues, respiratory dysfunction, and lower back pain (187). This is further corroborated by a 2012 study done by Willard et al. in which the fascia of the lumbar and thoracic areas of the spine is discussed in relation to its stabilization of the spine and pelvis (188).

For all of the reasons just listed, it is clear that fascia is not a passive wrapping but rather a dynamic and vital component in the body's support and movement structure.

❖❖ MUSCLES OF THE BODY

MUSCLE VIEW (ANTERIOR)

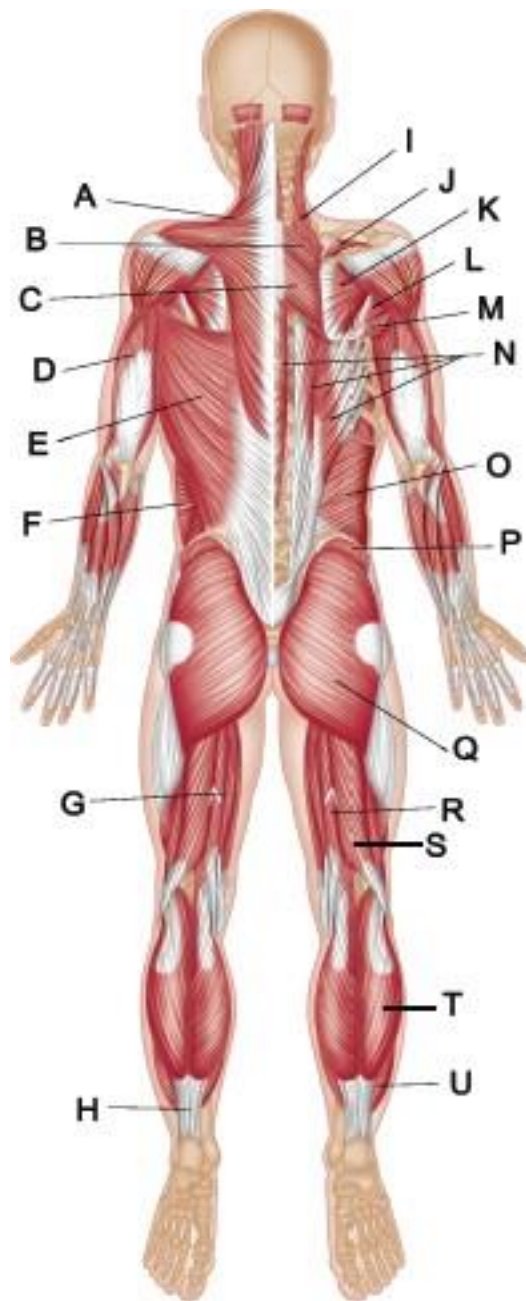


- A – Pectoralis Minor
- B – Deltoid
- C – Serratus Anterior
- D – Biceps Brachii
- E – Brachioradialis
- F – Pectineus
- G – Adductor Longus
- H – Rectus Femoris
- I – Vastus Medialis
- J – Vastus Lateralis
- K – Gastrocnemius
- L – Soleus
- M – Pectoralis Major
- N – External Oblique
- O – Internal Oblique
- P – Rectus Abdominis
- Q – Transverse Abdominis
- R – Tensor Fasciae Latae
- S – Sartorius
- T – Gracilis
- U – Peroneals
- V – Tibialis Anterior

Anterior View

Figure 3.5 Skeletal Muscles – Anterior

MUSCLE VIEW (POSTERIOR)



Posterior View

Figure 3.6 Skeletal Muscles – Posterior

Muscle Classification

Some of the muscles listed next could be classified in multiple categories, such as the hamstrings (core and lower body). However, for our purposes, the category in which a muscle is listed here is based on its significance within that area. For example, while the hamstrings could be listed as “lower body” muscles, they are listed as “core” muscles because of their role in stabilizing the lumbar spine/pelvic/hip region. Additionally, not all muscles of the body are noted in the certification.

CORE MUSCULATURE

The core is perhaps the most overhyped, overmarketed, and least understood area of the body. When most people refer to the core, they think of six-pack abs and that the more “ripped” one’s abs are, the stronger the core is. This could not be further from the truth.

Like other areas of exercise science, one of the primary issues in defining the core is that there are multiple definitions, many of which are driven by product marketing and a lack of knowledge.

Within this certification, when the term “inner core” is used in relation to abdominal, oblique, and back muscles, it refers to deep muscles, meaning muscles that are at least one layer below the surface muscles that most people refer to as the core. In other words, if you can see a muscle (i.e., superficial), it is not a primary core muscle.

The core area is also commonly referred to as the *lumbo-pelvic-hip complex* (LPHC). This term is more appropriate than simply “core,” as it specifically addresses the area that stabilizes the upper and lower body. Therefore, muscles that act to stabilize this complex can be viewed as core musculature. The LPHC encompasses both the inner (primary) and outer unit core muscles.

The primary core muscles are:

- **Multifidus (lumbar)**
- **Pelvic Floor Muscles**

- **Transverse Abdominis (TVA)**
- **Internal Oblique**
- **Diaphragm**

A good visual cue for the core (primary) muscles is to imagine a box (147). In order for a box to be strong, all of its sides must provide stability.

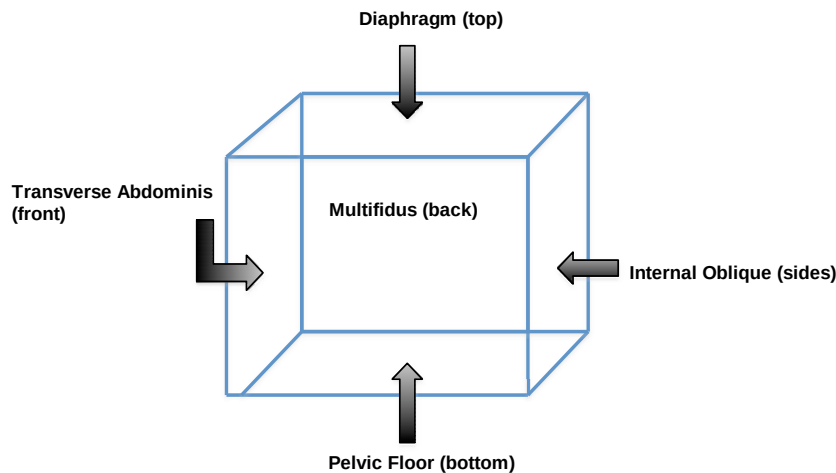


Figure 3.7 Primary (Inner) Core Musculature Diagram

As a point of reference, the muscle that most people consider the core is the *rectus abdominis*. The *rectus abdominis* is the “six-pack” muscle. This is a superficial muscle, and while it does provide some stability to the LPHC, its primary job is anterior spinal flexion.

The primary purpose of the core musculature is to provide stability and rigidity to the area of the spine and pelvic region. The force production of the core muscles is not great, as their primary function is to create stability, not gross movement. Synergy between the core musculature and ligaments allows for substantial stability.

Given the confusion over exactly what muscle or muscles constitute the core, it is advised to separate the *core* into two areas: *inner unit* and *outer unit*.

The inner unit is made up of the muscles just described as *primary core muscles*. The outer unit is made up of the following muscles:

- **Rectus Abdominis**

- External Obliques
- Erector Spinae
- Quadratus Lumborum
- Iliopsoas (made up of Iliacus and Psoas)
- Rectus Femoris
- Hip Adductors
- Glute Maximus
- Hamstrings

While the outer unit muscles provide stabilization to the LPHC, they are not as specialized to this task as the inner unit. As noted earlier, the primary function of the outer unit muscles is to provide movement, not stabilization. Therefore, when the outer unit muscles become overactive, they essentially shut off the inner unit musculature. This greatly reduces the amount of LPHC stabilization. The inner unit is the primary LPHC stabilizer, and the outer unit is the secondary stabilizer.

LATERAL SCHEMATIC DIAGRAM OF INNER UNIT CORE MUSCULATURE

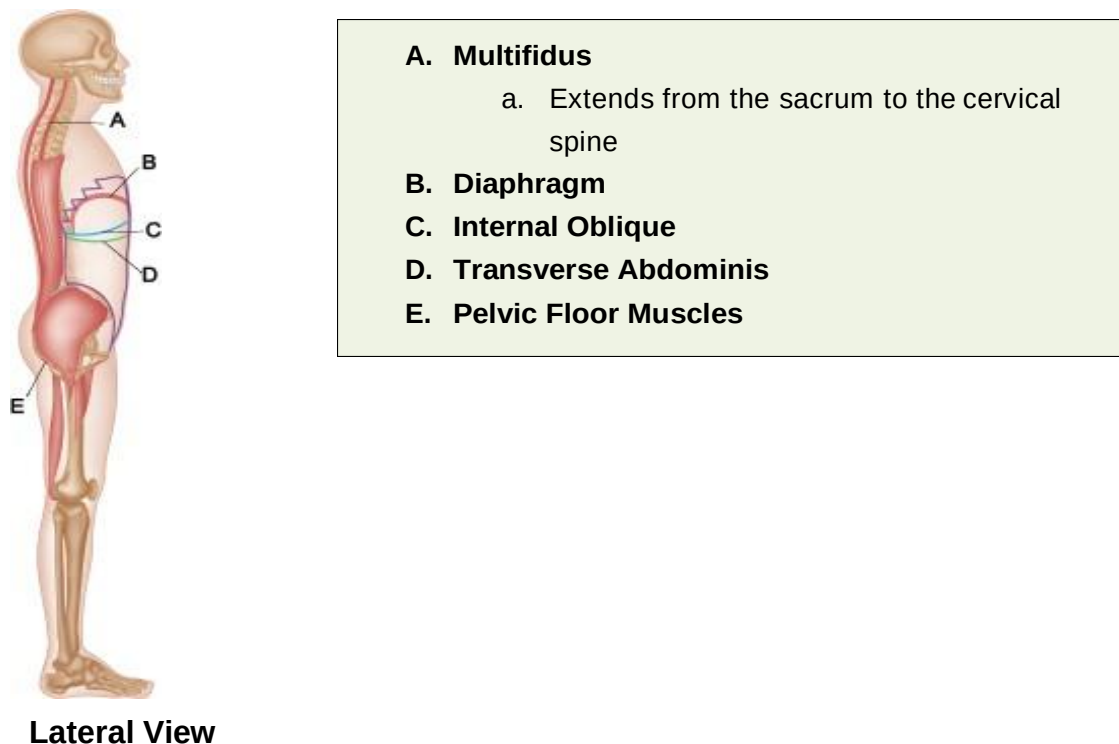


Figure 3.8 Primary (Inner) Core Musculature Landmarks – Lateral View

As most people view the outer unit as the core, it is common for people to overwork their outer unit muscles, which as just noted can result in the shutting down of the inner unit core musculature. Individuals who perform countless abdominal crunches and leg raises believe they are strengthening their core muscles, and thus increasing their lumbar support, when, in fact, the opposite is often the case.

The inner unit, and more specifically the TVA, stabilizes the lumbar spine to allow both upper and lower body motion by contracting 90 milliseconds prior to upper-body motion and 120 milliseconds prior to lower-body motion (146). This illustrates the importance of the TVA in providing stability to the lumbar spine.

Inner Unit Core Musculature



INTERNAL OBLIQUE

This muscle can be considered the intermediate muscle of the core because of its placement, as it lies between the transverse abdominis and the external oblique. In addition to providing spinal stability, it has two additional functions. First, the contraction/relaxation of the internal oblique results in movement of the diaphragm, which is responsible for inhalation and exhalation. Second, the internal oblique is responsible for rotation and lateral flexion (side-to-side bending) of the upper body.

Lateral View

Figure 3.9 Internal Oblique



Lateral View

Figure 3.10 Transverse Abdominis

TRANSVERSE ABDOMINIS (TVA)

This is the deepest of all the abdominal muscles, and its primary action is to provide lumbar stability. The TVA acts as the body's natural weight belt by providing hoop tension around the midsection of an individual.



Anterior View

Figure 3.11 Diaphragm

DIAPHRAGM

The diaphragm is a “domelike” sheet of muscle located between the thoracic and abdominal cavities and is responsible for the increase and decrease of lung size when inhaling (contraction of diaphragm) and exhaling (relaxation of diaphragm), respectively. The diaphragm also provides lumbar stability by working with the other inner unit core muscles.



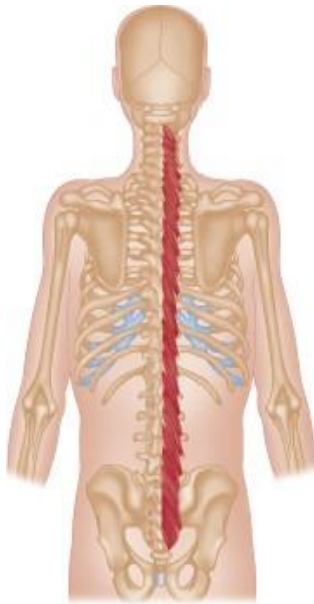
Anterior View

Figure 3.12 Pelvic Floor Muscles

PELVIC FLOOR MUSCLES (PF)

Comprising the *levator ani*, *coccygeus*, and connective tissue, the pelvic floor muscles essentially form a “hammock” that supports pelvic organs (e.g., bladder). The PF muscles also assist in bladder and bowel function. Specifically, in relation to core musculature, the PF muscles assist in stabilization of the spine and pelvis.

A: Pelvic Floor Muscles



Posterior View

Figure 3.13 Multifidus

MULTIFIDUS (Lumbar)

The multifidus (mull-tiff-a-duss) is actually a series of small muscles that attach laterally to both sides of the spinal column and is deep to the erector spinae. The multifidus is broken down into segments with the insertion point several vertebrae higher than the origin. This segmental construction means that each multifidus segment is responsible for controlling a small section of the spine.

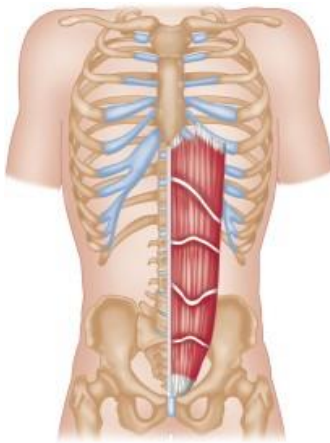
The multifidus can be broken down into superficial and deep muscles. The superficial multifidus is responsible for lumbar back extension, rotation, and lateral flexion while the deep multifidus is responsible for spinal stabilization.

The correlation between the strength (spinal stability) of the multifidus and lower back pain was documented in a 2002 study by Danneels et al. published in the *European Spine Journal*. In this study, a group of individuals with healthy backs were compared with a group that suffered from lower back pain. It was demonstrated via EMG testing of the multifidus that the group with lower back pain demonstrated much lower multifidus activation than the healthy group. Additionally, the lower back pain group had reduced ability to voluntarily contract their multifidus over the healthy group (145).

The multifidus co-contracts with the TVA to provide spinal stability. Like the TVA, the multifidus contracts before most body movements as a protective mechanism for the spine.

Outer Core Musculature

The primary purpose of the outer core musculature is to produce movement and, secondarily, to assist in providing stabilization to the LPHC.

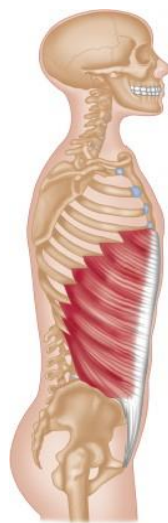


Anterior View

Figure 3.14 Rectus Abdominis (left side)

RECTUS ABDOMINIS

This muscle group is made up of two muscles that run parallel to each other and are separated by connective tissue (*linea alba*). The rectus abdominis is primarily responsible for flexion of the lumbar spine. Secondarily, the rectus abdominis assists in stabilizing the lumbar spine.

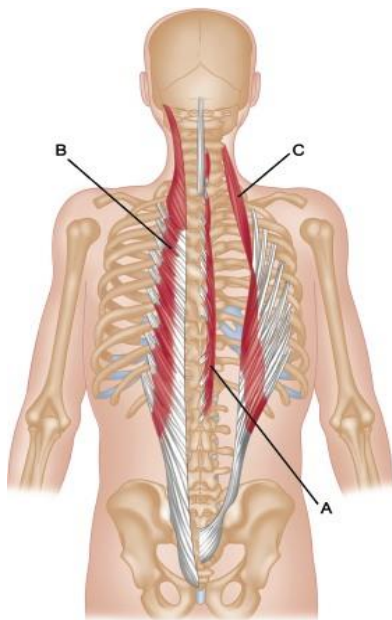


Lateral View

Figure 3.15 External Oblique

EXTERNAL OBLIQUE

This muscle is located on either side of the rectus abdominis and is responsible for flexion of the spine as well as rotational and lateral flexion (bending side to side) of the spine. Like the rectus abdominis, it secondarily assists in stabilizing the lumbar spine.



Posterior View

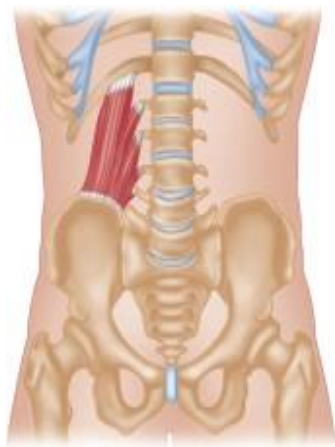
Figure 3.16 Erector Spinae

ERECTOR SPINAE

Also called *extensor spinae*, this group of muscles runs parallel to the vertebral column from its base (lumbar) through the top (cervical) section. The three groups of muscles are:

- A. Spinalis (Medial)**
- B. Longissimus (Center)**
- C. Iliocostalis (Lateral)**

The primary function is extension and lateral flexion of the spine. Secondly, it provides stabilization to the spine.

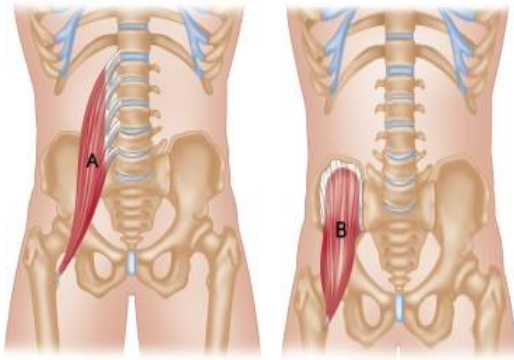


Anterior View

Figure 3.17 Quadratus Lumborum

QUADRATUS LUMBORUM (QL)

Originates at the iliac crest and inserts into the lowest rib and side of the lumbar vertebrae. The QL acts to stabilize the pelvis and laterally flex the lumbar spine. If the glute medius is underactive, the QL can initiate hip abduction, which often leads to the QL becoming overactive. Rotation of the pelvis also can result in the QL becoming overactive.



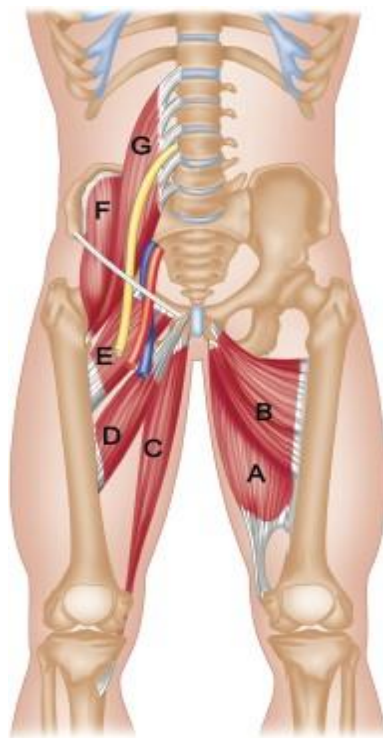
Anterior View

Figure 3.18 Hip Flexors

HIP FLEXORS

The psoas (so-as) and iliacus (ill-e-ack-us) make up a group of muscles commonly termed *hip flexors*, due to their primary role.

- A. Psoas**
- B. Iliacus**



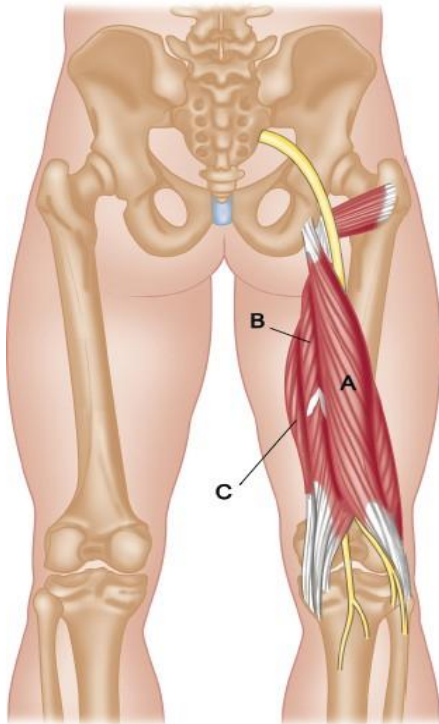
Anterior View

Figure 3.19 Hip Adductors

HIP ADDUCTORS

Below are the muscles responsible for hip adduction:

- A. Adductor Magnus**
- B. Adductor Brevis**
- C. Gracilis**
- D. Adductor Longus**
- E. Pectineus**
- F. Iliacus**
- G. Psoas Major**



Posterior View

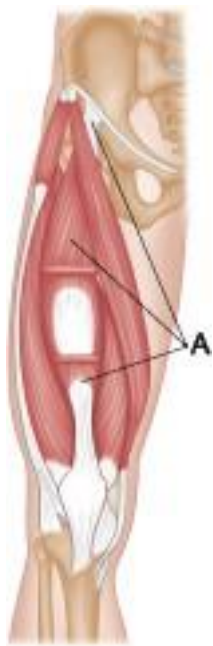
Figure 3.20 Hamstrings

HAMSTRINGS

The hamstrings comprise three muscles:

- A. Biceps Femoris**
- B. Semitendinosus**
- C. Semimembranosus**

These muscles are used to stabilize the LPHC, flex the lower leg, and extend (posterior) the femur. During the gait cycle, flexion of knee is largely passive.



Anterior View

Figure 3.21 Rectus Femoris

RECTUS FEMORIS

The rectus femoris is one of four muscles that make up the quadriceps and is visually the center muscle of the four when viewing the muscle in the frontal plane. The primary role of the rectus femoris is knee extension and hip flexion. However, it also plays a role in stabilizing the LPHC and the knee.

- A. Rectus Femoris (cut)**



Posterior View

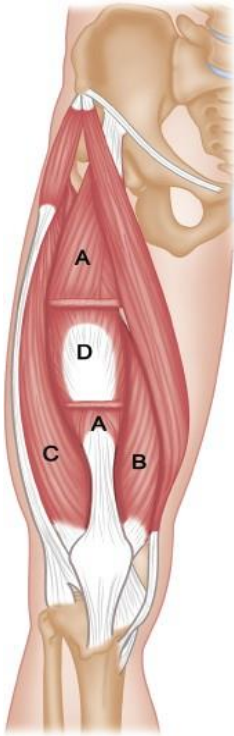
Figure 3.22 Gluteus Maximus

GLUTEUS MAXIMUS

The gluteus maximus is the most superficial of the three gluteal muscles (gluteus medius, gluteus minimus). Its primary function is support of the LPHC and extension/external rotation of the hip.

A. Gluteus Maximus

LOWER-BODY MUSCULATURE



Anterior View

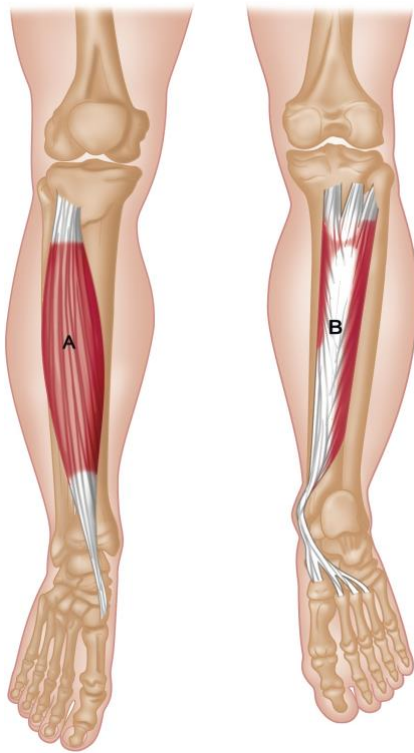
Figure 3.23 Quadriceps

QUADRICEPS

The “quads,” as this muscle group is often called, consist of four different muscles and are located on the anterior side of the upper leg.

- A. Rectus Femoris (cut)**
- B. Vastus Medialis**
- C. Vastus Lateralis**
- D. Vastus Intermedius**

These muscles are used to extend the lower leg and stabilize the knee. The rectus femoris also acts to flex the hip, as noted previously in the *Outer Core Musculature* section.



Anterior View

Posterior View

Figure 3.24 a) Tibialis Anterior b) Tibialis Posterior

FOOT FLEXORS/EXTENSORS (Calves)

A. Tibialis Anterior: located in front of the tibia. Action: dorsiflexion and foot inversion

B. Tibialis Posterior: deep muscle, located behind tibia. Action: plantar flexion, foot inversion and stabilization



Posterior View

Figure 3.25 Soleus

Soleus: deep muscle, located behind tibia and underneath the gastrocnemius. Action: plantar flexion and stabilization

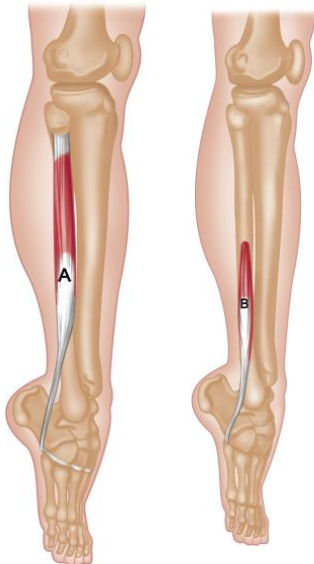


Posterior View

Figure 3.26 Gastrocnemius

Gastrocnemius: superficial muscle, located on top of soleus. Action: plantar flexion and stabilization

- The term ***triceps surae*** (three heads) is often used to refer to both the gastrocnemius and the soleus.



Lateral View

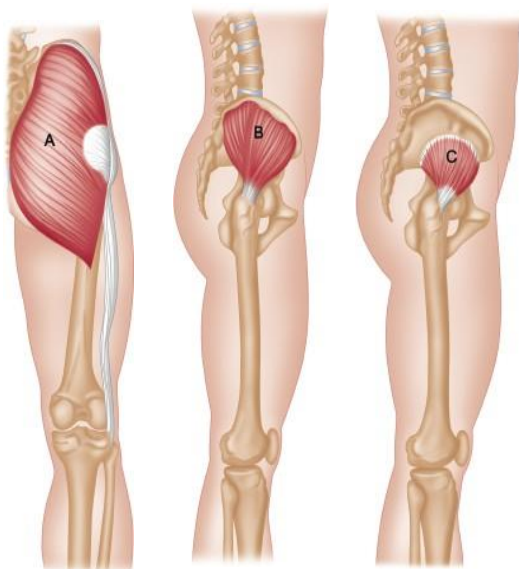
Figure 3.27 Peroneals

Peroneals: These muscles are on the lateral aspect of the lower leg.

A. Peroneous Longus

B. Peroneous Brevis

Action: plantar flexion and foot eversion



Lateral View

Figure 3.28 Gluteals

GLUTEALS

Commonly referred to as the *glutes*. Three muscles make up the gluteals:

- A. Gluteus Maximus** (previously noted in *Outer Core Musculature*)
- B. Gluteus Medius**
- C. Gluteus Minimus**

The gluteus minimus and medius are responsible for abduction and extension, as well as internal rotation (when hip is extended) and external rotation (when hip is flexed) of the hip. These muscles also stabilize the pelvis.



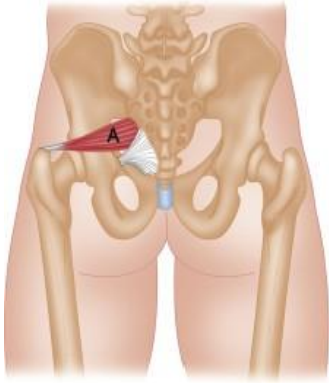
Lateral View

Figure 3.29 Tensor Fasciae Latae

TENSOR FASCIAE LATAE

In addition to the gluteals, the *tensor fasciae latae* (commonly referred to as the TFL) also acts to abduct the hip. The TFL also extends and externally rotates the hips. The TFL is located at the side of the hip

- A. Tensor Fasciae Latae**



Posterior View

Figure 3.30 Piriformis

PIRIFORMIS

Abducts the femur when the hip is flexed. For many individuals, the sciatic nerve passes through the piriformis. When the piriformis tightens, the sciatic nerve may become compressed and induce sciatica (pain that originates at the gluteals and often travels down the leg via the sciatic nerve). This is also called *piriformis syndrome*.



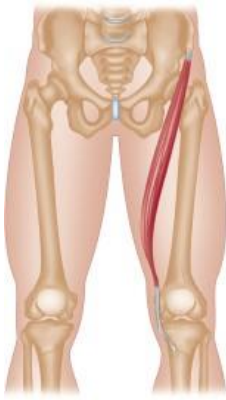
Posterior View

Figure 3.31 Popliteus

POPLITEUS

This is a relatively small muscle located behind the knee. The primary action is to flex the knee, and more specifically, it is responsible for “unlocking” the knee from a straightened position. It also assists with internal rotation of the tibia.

Overpronation of the foot or hyperextension of the knee during running can injure the popliteus. This injury often presents with difficulty and/or pain when extending the knee.



Anterior View

Figure 3.32 Sartorius

SARTORIUS

This is the longest muscle in the body and begins (originates) below the ASIS and ends (insertion) on the superior and medial aspect of the tibia. This muscle acts to flex, abduct, and externally rotate the hip. It also aids in knee flexion.

UPPER-BODY MUSCULATURE

Chest Musculature

The two primary muscles that make up the chest are:



Anterior View

Figure 3.33 Pectoralis Major

PECTORALIS MAJOR

Responsible for adduction and internal rotation of the humerus.



Anterior View

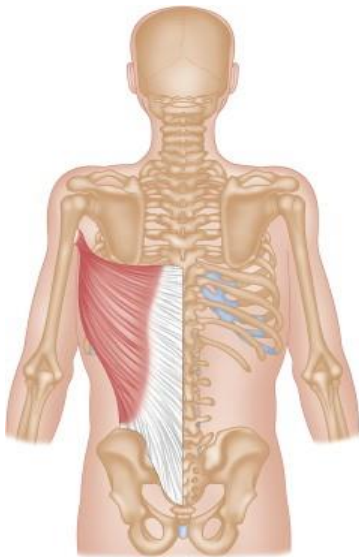
Figure 3.34 Pectoralis Minor

PECTORALIS MINOR

Moves the scapula forward and downward. Located underneath the pectoralis major.

Back Musculature

This certification will focus on five primary muscles of the back.

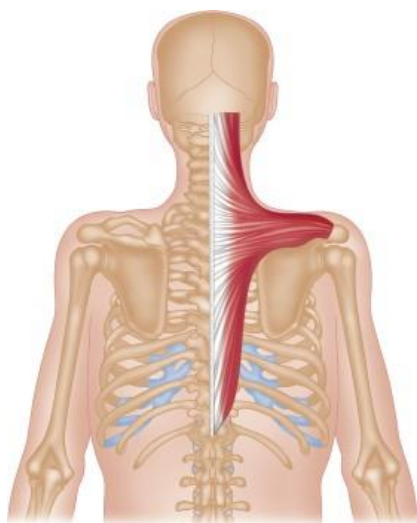


Posterior View

Figure 3.35 Latissimus Dorsi

LATISSIMUS DORSI (Lats):

Largest muscle in the back and primarily responsible for adduction, extension, horizontal abduction, and internal rotation of the shoulder joint. It also assists in stabilizing the LPHC.

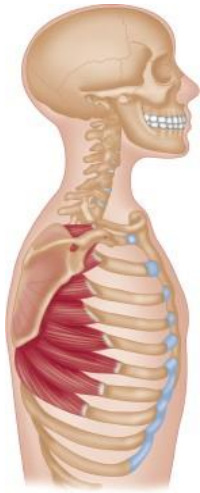


Posterior View

Figure 3.36 Trapezius

TRAPEZIUS

Large, diamond-shaped muscle located on either side of the midupper spine. The primary action of the trapezius is to move the scapula and provide support to the upper arm. The trapezius stabilizes, elevates, rotates, depresses, and retracts the scapula. The upper trapezius produces cervical extension, lateral flexion, and rotation.

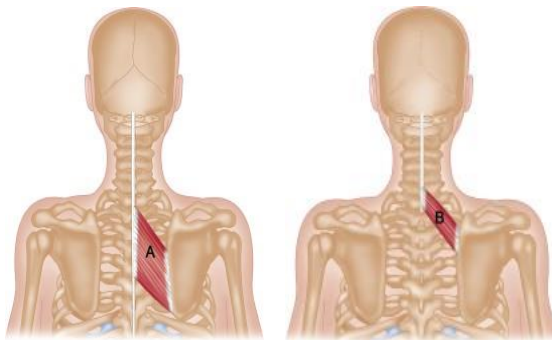


Lateral View

Figure 3.37 Serratus Anterior

SERRATUS ANTERIOR

Originates at the ribs and attaches to the medial side of the scapula. Its function is to protract, stabilize, and provide upward rotation of the scapula. This muscle is often termed the *boxer's muscle* as it pulls the scapula forward, which occurs when throwing a punch.



Posterior View

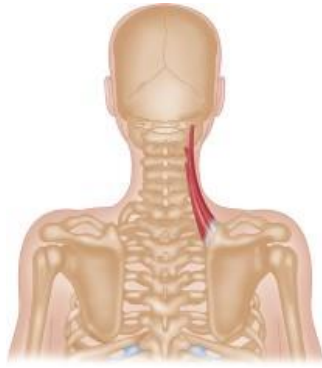
Figure 3.38 Rhomboids

RHOMBOIDS

Originate on the spine and attach to the scapula. They stabilize, retract, and rotate the scapula.

A. Rhomboid Major

B. Rhomboid Minor



Posterior View

Figure 3.39 Levator Scapulae

LEVATOR SCAPULAE

Elevates scapulas. Assists in scapular retraction and lateral flexion and extension of the neck.

Shoulders (Deltoids)

Many muscles act on the shoulder (*glenohumeral*) joint. For the purpose of this certification, we will focus on the deltoid muscles (anterior, posterior, and lateral) and the **rotator cuff** muscles, which are made up of the *supraspinatus*, *infraspinatus*, *teres minor*, and *subscapularis*. The rotator cuff muscles act to provide static and dynamic support to the glenohumeral joint (159). The actions of the four rotator cuff muscles are noted below:



Lateral View

Figure 3.40 Deltoids

DELTOID (Shoulder) MUSCLES

- ☐ Lateral
- ☐ Anterior
- ☐ Posterior



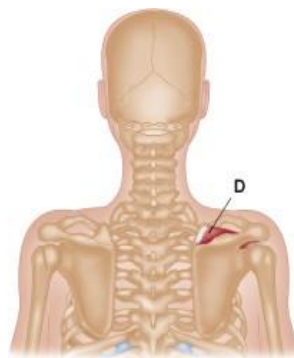
A: Posterior View



B: Posterior View



C: Anterior View



D: Posterior View

ROTATOR CUFF MUSCLES

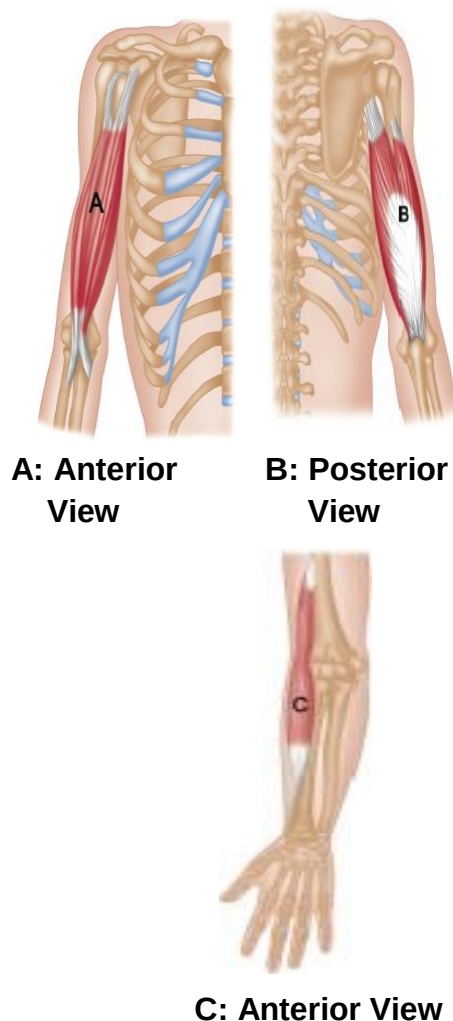
- A. Infraspinatus:** external shoulder rotation
- B. Teres Minor:** external shoulder rotation
- C. Subscapularis:** internal shoulder rotation
- D. Supraspinatus:** shoulder abduction

The deltoid and rotator cuff muscles provide mobility and stability to the glenohumeral joint. The rhomboids and trapezius provide mobility and stability to the scapula.

Figure 3.41 Rotator Cuff: a) Infraspinatus b) Teres Minor c) Subscapularis d) Supraspinatus

Arms

While running, the arms assist in providing stabilization and balance.



PRIMARY ARM MUSCLES

- A. Biceps Brachii:** elbow flexion
- B. Triceps Brachii:** elbow extension
- C. Brachioradialis:** wrist extension

In addition to these muscles are over 20 muscles that make up the forearm that act to move the wrist, forearm, and digits (fingers). While not all the forearm muscles will be mentioned in this certification, it is important to know that these muscles are responsible for forearm rotation as well as wrist and digit extension/flexion.

Figure 3.42 Primary Arm Muscles:

a) Biceps Brachii b) Triceps Brachii c) Brachioradialis

❖❖ INTRAMUSCULAR PERFORMANCE

WARM-UP

For shorter distance events such as 10K and below, a proper warm-up is a necessity because of the short distance and intensity of the race. A warm-up is also advised for longer events and training runs. The primary purpose of a warm-up is to increase one's core temperature (625). An increase in core body temperature positively affects the biomechanics of a runner as well as decreases the chance of injury such as muscle strains (625, 626).

Below are several areas that a proper warm-up enhances:

- ☐ Increased force capacity of a muscle (627)
- ☐ Decreased chance for injury (626)
- ☐ Increased range of motion (628)

Stretching vs. Warm-up: Are They the Same?

It is common for stretching and warming up to be lumped into the same category. This is often the case as athletes are commonly instructed to warm up their muscle(s) prior to stretching to reduce the chance for injury while stretching. In this scenario, both can be considered part of the warm-up. However, as the name suggests, a warm-up primarily relates to warming the body, which therefore includes increasing the temperature of muscles.

When researched in isolation, a warm-up reduced the incidence of lower-limb injury whereas stretching alone did not (629). The primary reason for stretching is to increase range of motion around a joint. This is pertinent, as research has shown that stretching a muscle prior to a bout of exercise can reduce the contractile force of a muscle (608).

In other words, stretching can be part of a warm-up routine but it is not advised to do in isolation.

ACTIVE vs. PASSIVE WARM-UP

An active warm-up occurs when the changes in body temperature are due to muscle activity (i.e., running), whereas a passive warm-up is due to an external source (i.e., heating pad). It is advised to perform an active warm-up rather than a passive warm-up.

A **Related Warm-Up** is a subset of an active warm-up and denotes warming up (via active warm-up) the same muscles used in the activity being performed and, ideally, in the same range of motion. This is the advised type of warm-up.

Warm-Up Guidelines

The goal of a proper active warm-up is to find the sweet spot between the muscles becoming sufficiently warm but not physically taxing the individual. As a result, a proper active warm-up can be difficult to prescribe and perform.

Generally speaking, the more conditioned an individual is from a cardiovascular standpoint, the longer the warm-up will be. Research by Stewart and Sleivert found that an active warm-up of 15 minutes and at an intensity of approximately 65 percent of one's VO_2 max was most beneficial in regard to anaerobic activities (631). Another study by Tomares et al. found that in regard to sprint track cycling, shorter and lower-intensity warm-ups elicited better performance than longer warm-ups (>20 min of cycling at varying intensities) (632). While these studies primarily examined a warm-up's effect on anaerobic activity, it seems to demonstrate that a long warm-up (>20 minutes) may have detrimental effects on an individual's energy and power output.

It is strongly recommended to try out different warm-up protocols in training to determine what protocol elicits the best result for your client.

Perhaps the most important finding regarding warming up is not the exact intensity or duration of the warm-up but rather the time duration between the cessation of the warm-up and the start of the competition. The benefits gained via an increase in muscle temperature are lost approximately 15 minutes after the warm-up period ends. Therefore the time between the end of the warm-up and the start of the competition should be no longer than 15 minutes (633). While not discussed, it stands to reason that the environmental temperature would

impact the desired window between warm-up and event start (i.e., cold weather = shorter time period than 15 minutes).

Regardless of the warm-up duration and activity, muscles work better when they are warm versus cold. More specifically, muscles work optimally when the body core temperature is at approximately 100.4°F and a muscle temperature of 101.3°F. However, like most aspects of human performance, the temperature at which muscles work optimally exists on a bell curve. For example, the contractile force of a muscle decreases once the muscle temperature goes above 103.3°F (760).

DELAYED ONSET MUSCLE SORENESS (DOMS)

While there may be muscle soreness and weakness directly after a workout or a bit the next morning, the most severe pain, or DOMS, usually occurs within 24–48 hours following strenuous activity. For many years DOMS was a mystery and not fully understood. Many studies have been published on this subject, yet most did not fully explain what DOMS really is. In 1981, Jan Frieden published his findings that were based on taking muscle biopsies from subjects who walked repeatedly down flights of stairs. This was the first real evidence of what DOMS is and what it is not (23).

Jan Frieden discovered that DOMS primarily results from eccentric contractions of muscles (23). An eccentric contraction is when a muscle contracts to decelerate a limb that has a load applied to it. Essentially, an eccentrically contracting muscle acts as a brake. For example, after performing a bicep curl, the bicep muscle must contract to decelerate the weight on the way back down to the starting position (arm extended downward). During an eccentric contraction, the muscle fibers get pulled lengthwise to the point where damage occurs at a subcellular level. This damage induces an inflammatory response of a muscle (260). The amazing thing about the body is that, given the proper amount of rest, the damaged muscles rebuild stronger than they were before.

Additionally, eccentric contractions involve fewer motor units than concentric contractions to elicit the same force production. Therefore during an eccentric contraction, there is greater stress per muscle fiber than during a concentric muscle contraction. This added stress per muscle fiber may contribute to DOMS because of disruption of the myofibrillar structures of the muscle (634).

While many people view DOMS as a bad thing, in some respects it is actually a good thing. The pain that people feel with DOMS is, in fact, the muscle rebuilding rather than breaking down. One way to look at DOMS is to relate it to the flu. While most people view the side effects of the flu as negative because they feel horrible because of high body temperature, chills, and aches, the side effects of the flu represent the recovery phase. The high temperature is the body's protective response to infection and injury. Just as one would not be active when having a temperature, one should also not stress the same muscles affected by DOMS while they are sore. As the soreness is representative of the healing phase, if you stress the muscles affected by DOMS, this will inhibit healing and further break down the muscle fibers. Many athletes and coaches subscribe to the "no pain, no gain" philosophy. However, when it comes to DOMS, if you try to push through DOMS by exercising the affected muscles, you will end up doing more harm than good.

DOMS is not limited to strength training. Any time a muscle is stressed to a level at which it is not adapted to, damage can occur. DOMS is often the resulting aftereffect. Runners who increase their mileage too much on a particular day will often feel the effects of DOMS.

ROLE OF ESTROGEN ON DOMS

From a biochemical standpoint, a marker for muscle damage (i.e., DOMS) is the enzyme creatine kinase (CK) in the blood. Research has shown that estrogen may affect muscle enzymes that result in lower blood CK levels. Therefore it can be theorized that women may have less severe bouts of DOMS than men.

Cause of Muscle Soreness and Weakness

The prevailing thought is that DOMS-related muscle weakness is due to subcellular muscle damage. However, research has discovered that undamaged muscle fibers can exhibit muscle weakness (635). Therefore, muscle weakness is likely due to a muscle's inability to activate contractile structures due to the overstretching of sarcomeres (636).

As the time course of inflammation is typically the same as for muscle soreness, the soreness that is felt during DOMS is likely attributable to the inflammation response (637). As noted above, the overstretching of the sarcomeres likely creates a loss of cellular calcium homeostasis, which is reason for the muscle soreness (638).

Is a Cooldown Necessary?

A cooldown is characterized by a large reduction in intensity after an exercise session is over. The popular purported benefit of cooling down is to bring down the heart rate in a steady fashion while allowing the muscles to gradually return to normal tension levels. Is all this really necessary?

It is theorized that warming up before exercise can slightly reduce DOMS while a cooldown does not reduce DOMS (335). According to Dr. Hirofumi Tanaka, an exercise physiology professor at the University of Texas, the cooldown is largely a myth that like most myths in the fitness and exercise realms keeps getting passed along without much thought given to its validity (336).

The one subject all physiologists seem to agree on is that after abruptly ceasing intense exercise such as running, the blood vessels in the legs are dilated, and therefore, blood can pool quickly in the legs and feet. This can result in an individual feeling dizzy and potentially passing because of a lack of blood flow to the brain (336).

While this is a good reason to implement some form of cooldown, especially after running, there does not seem to be any other physiological reason (i.e., faster recovery) for one (336).

Therefore, one could deduce that lying down with the legs elevated to reduce or eliminate blood pooling would be just as, if not more, effective than a short cooldown after intense exercise.

Use of NSAID's to Counteract DOMS or Musculoskeletal Injury

Non-steroidal anti-inflammatory drugs (NSAID) are commonly taken for the purposes of pain reduction and to reduce inflammation. Drugs (and common associated brand names) that fall into this category are: *naproxen* (Aleve®), *ibuprofen* (Advil®, Motrin®), and *aspirin* (Bayer®).

Acetaminophen (Tylenol®) is also commonly used to reduce pain and fever. However, it is not classified as an NSAID because of its limited anti-inflammatory response.

Because of the pain-reducing and anti-inflammatory properties of NSAIDs, they are commonly used by those who experience DOMS and musculoskeletal injuries. However, research has shown that NSAIDs actually slow healing and muscle growth (133). The evaluation of healing speed was not just focused on muscle, but on bones and tendons as well.

Myosatellite cells, or satellite cells, are found in muscles, and their job is to attach to existing muscle fibers to form new fibers. This process occurs during normal muscle growth as well as during recovery from injury. It is theorized that NSAIDs greatly reduce the spread of these cells, thus hampering muscle recovery and growth (134).

Another issue in taking any sort of pain-reducing medication is that it allows an individual to potentially train past what the person's body can physically handle because of the medication-induced reduction of pain. This can lead to greater injury down the road. However, other research has deemed that short-term use of NSAIDs would have minimal impact on the healing process (135).

Conclusion

While you cannot advise or prescribe any type of medication to clients, you can inform them of the research in this area. While there is no definitive, agreed-upon effect of NSAIDs on muscle recovery, it would be wise to share this information with clients as the abundance of research to support the negative effect of NSAIDs on muscle recovery is too great to ignore.

MUSCLE BURN

For endurance athletes, one of the most feared byproducts of exertion is the dreaded *burn* that they feel in their muscles as the intensity and/or distance increases. So what exactly causes this burn? For many years (and still to this day for some people) the thought was that an increase in blood lactate within the body was the cause of the burning sensation. The primary reason for this belief was because of a study done in the 1920s by Dr. Otto Meyerhof. Using electric stimulation to a deceased frog's leg muscles, Dr. Meyerhof observed that the leg muscles of the frog twitched initially when subjected to the electric stimulation. However, after a while the twitching ceased. The cessation of muscle activation correlated with high levels of blood lactate. Thus, Dr. Meyerhof concluded that an increase in lactate causes muscular fatigue (24). This theory was challenged and ultimately put to bed by the research of Dr. George Brooks, who while studying for his PhD dissertation, found the opposite of what Dr. Meyerhof had proposed was true.

Dr. Brooks discovered that in actuality, blood lactate is an important muscle fuel and not a waste byproduct that causes the burn, as earlier thought (25). Pyruvate can be oxidized and used as fuel for the Krebs cycle, or it can be converted to lactate, which, in turn, is converted into glycogen in the liver via the Cori cycle. This increase in glycogen means that the glycogen stores are larger and therefore last longer.

What Is Lactic Acid?

If you've been around endurance sports for even a short time, you no doubt have heard of the term *lactic acid*. This is the product that most people associate with the *burn*. What people likely mean when they say lactic acid is actually *lactate*. Many people use these two terms interchangeably. However, they are actually two different compounds. Chemically speaking, lactate is lactic acid, minus one proton (30).

When a person begins to exercise, the oxygen consumption of the individual increases. However, at a certain point, an oxygen consumption plateau is reached. This plateau is often referred to as *steady state* or *steady rate*. What this means is that there is a balance between the energy being used by the muscles doing the work and the energy being created aerobically (more on this

topic in the next module). During this steady-state condition, pyruvate accumulation is minimal. This is because any pyruvate (an acidic byproduct of glycolysis) being produced is either oxidized or converted to lactate, which in turn is converted to glycogen, as previously mentioned (27).

Intramuscular levels of lactate can elevate rapidly if the intensity of exercise is extremely high. Research by Gaitanos et al. found that intramuscular levels of lactate can increase by 7 mM (a substantial increase) in just six seconds when performing high-intensity cycle efforts (559). Blood lactate levels increase to approximately 40 times their resting level when going from rest to maximal exercise intensity (566).

So, what actually causes the *burn*? The burn is the result of a buildup of acidity in the muscle cells. More specifically, this acidity is caused by the release of hydrogen ions during the fast turnover of ATP (26). This release of hydrogen ions within the muscle cells causes an interference with the ability of the muscle fibers to contract properly. Unless the intensity of the exercise decreases, the high level of burn (*acidosis*) will eventually cause cessation of the exercise (28). A role of lactate is to help buffer and neutralize the hydrogen ions, thereby delaying the burn (26).

While most athletes view the burning sensation as a bad thing, it is the body's way of self-regulation so that it does not push itself too far and cause damage. The burning sensation is akin to a governor in a car motor that limits how fast the car can go. The theory of a governor is discussed in the *Pacing* module.

Lactate threshold is discussed in *Module 5: Training Benchmarks*.

EXERCISE-ASSOCIATED MUSCLE CRAMPS

Previously thought to be the result of a lack of hydration and/or electrolytes, the most recent research seems to indicate that exercise-associated muscle cramps (EAMC) are primarily the result of a muscle or muscles being overworked (i.e., fatigue). Meaning, if a muscle has a substantially greater load applied to it that requires a faster or stronger contraction than it is used to, or if a muscle is activated for longer durations of time without rest than it is used to, cramping is often the result.

For example, if you never do squats as an exercise and you proceed to do so with a heavy weight, it is likely that cramps may result. Were you dehydrated? Likely no. Were your electrolyte levels low? Probably not.

According to Dr. Martin Schwellnus of the University of Cape Town, EAMC is the result of an abnormality of neuromuscular control that is in response to muscle fatigue – not a depletion of electrolytes (758).

In a 2011 study by Schwellnus et al., 210 triathletes were recruited to examine risk factors for EAMC (759). Of the 210 triathletes, 43 reported suffering from EAMC while 166 did not experience EAMC. Those who suffered from EAMC predicted faster race times and had faster times than those who did not suffer from EAMC. This is important to note because the two groups (EAMC, non-EAMC) had similar prerace training preparation and performance histories. Lastly, the pre- and post-race blood serum electrolyte concentrations, as well as body weight, were the same in both groups. These findings suggest that dehydration and electrolyte balance are not a cause for EAMC, whereas exercise intensity is a major contributing factor.

❖❖ MUSCLE SYNERGY

Even the simplest muscle movement requires synergy between muscles in order for movement to occur. From a macro point of view, the human body is made up of many opposing muscles. Let us use the elbow as an example. The bicep muscle is responsible for flexing the elbow, whereas the tricep muscle is responsible for extending the elbow. When one muscle is shortening, the opposing muscle is lengthening, and vice versa. While this example illustrates only one movement of one joint, when an action requires a multijoint, multiplanar movement, the synergy between muscles is substantially more complex.

In order for a runner to be as efficient as possible, the relationships between working muscles need to be as synergistic as possible. When this synergy breaks down for any reason, performance is diminished and the potential for injury dramatically increases.

KINETIC LINKING

The term *kinetic linking* (also referred to as the *kinetic chain*) corresponds to the order in which muscles activate and work together to allow the body to move. From a performance-based standpoint, it also pertains to how muscles work to give an individual a biomechanical advantage. An example of kinetic linking is a boxer throwing a punch. If a boxer were to stand with feet in the frontal plane and throw a punch only from the shoulder, the punch would be quite weak and lack range. However, if the boxer utilizes kinetic linking, the hierarchy would look something like this:

1. Feet positioned diagonally to provide a strong base of support
2. Push body weight onto the ball of the trail foot
3. As the trail foot lifts from the heel onto the ball of the foot, the hips rotate
4. The arm begins to come forward
5. End position: hips are rotated in the transverse plane and the arm is fully extended forward

While this process is broken down into five areas for explanation purposes, in reality all five actions occur simultaneously. Putting the weight on the trail leg gives the boxer a strong base of support from which to push off. The shifting of the weight to the ball of the foot allows the hips to rotate more so than if the foot stayed flat. Rotating the hips allows for not only more momentum but also a greater reach, which equals a faster, more powerful punch. The longer the arm can travel, the greater the speed at which the punch can land, and the more substantial the impact.

Kinetic linking is all about optimizing performance. The goal is to be able to have the body move in the correct pattern to offer a biomechanical advantage. This requires the muscles to fire in the proper sequence so movement can occur seamlessly and in the correct position(s).

In functional body movement, muscles and body parts do not work in isolation. Consequently, incorrect movement patterns in one area of the body may be caused by dysfunction in another area of the body.

Effect of Glute Medius Weakness

A common issue that many runners have is iliotibial band syndrome (ITBS). The IT band originates at the top of the pelvis and runs down the side of the femur and inserts on the lateral side of the top of the tibia. Pain from ITBS is caused by an overtight IT band rubbing against the *lateral femoral epicondyle* (lateral side protrusion at the base of femur) when the knee flexes and extends. This results in pain on the lateral side of the knee. So what does this have to do with the kinetic chain?

While there are several potential causes of ITBS, a common cause is a tilted pelvis in the frontal plane. In this scenario, if a runner's gluteus medius is weak, it may cause the pelvis to tilt downward to the opposing side. This tilt results in the IT band on the side of the weak glute medius to become elongated (i.e., tighter) in relation to the opposing IT band – therefore causing ITBS. Further up the kinetic chain, the shoulders often tilt to compensate for the pelvic tilt, which therefore causes the head to tilt, and thus a whole host of muscle imbalances and likely pain are the result.

Additionally, when the glute medius is weak, the quadratus lumborum on the opposing side often becomes overactive in an attempt to stabilize the pelvis since the glute medius isn't assisting in pelvic stabilization. This often causes low back pain.

The following image denotes someone with correct posture (A) and someone with compensated posture due to a weak glute medius (B), as just described.

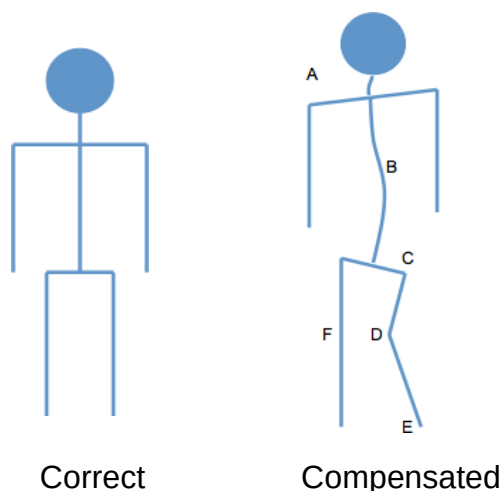


Figure 3.43 Correct versus compensated posture

- A: Tilted shoulders
- B: Curved spine
- C: Tilted pelvis (due to weak glute medius)
- D: Internally rotated femur
- E: Foot arch collapses
- F: Tightened IT band due to elevated pelvis

While figure 3.43 illustrates a person in a nonrunning position, the same compensations are visible when running.

This is an example of how muscle weakness or imbalance in one part of the body affects one or more parts of the body.

MUSCLE IMBALANCES

The question is not so much if a client has muscle imbalances but rather which muscles are imbalanced and to what degree. Muscle imbalances can result in decreased performance and, often, injury.

A muscular imbalance exists when there is a lack of balance between two opposing muscles that surround a joint. An example of a muscle imbalance would be if the upper trapezius was tighter than the mid and lower trapezius. This would likely lead to elevation of the shoulders.

Muscle imbalances can be caused by many factors, including habitual and repetitive movements, incorrect posture, restricted fascia, injuries, or surgeries. Muscle imbalances affect a runner in the following ways, all of which negatively influence performance:

- ☐ Improper biomechanics
- ☐ Potential for degeneration of joints
- ☐ Overuse of muscles, which can lead to injury
- ☐ Premature muscle fatigue

One way to illustrate a muscle imbalance is to think of the game of tug-of-war, with each team representing opposing muscles. In a perfect world, both teams would be evenly matched and there would be no movement of the rope. However, when a muscle imbalance is present, one team (i.e., muscle) is



Figure 3.44 Muscle synergy demonstrated via tug-of-war

Tug of war de: Tauziehen {{PD-USGov-Military-Navy}}
http://www.navy.mil/view_single.asp?id=24509 U.S. Naval Academy,
 Annapolis, Md. (May 17, 2005)

stronger than the other and will pull the rope – and the other team – to its side. For this imbalance to be corrected or minimized, the weaker team must become stronger to match the strength of the opposing team, or the stronger team must reduce its pulling force to equal that of the weaker team.

The term *shut down* is often used to denote muscles that contract minimally or not at all. Using running stride as an example, when the foot hits the

ground in the drive phase, the glutes and hamstrings should fire. For athletes whose glutes are shut down, the glute muscles are not recruited to the degree that they should be. Therefore the hamstrings are forced to take over the role of the glutes. In essence, the hamstrings are performing two jobs at once. This can lead to muscle strains and tendonitis of the hamstrings. If your client is unable to activate muscles through training or often incurs hamstring injuries, it is advised to refer the person to a physical therapist.

Virtually no part of the body works in isolation. One way to think about this is to compare a small muscle imbalance to a crack in the foundation of a house. If the crack is not repaired, it could go from a relatively minor repair to a major problem!

Length/Tension Relationship

A major determinant of a muscle's force production has to do with its length. When the word *length* is used in this section, it has to do with the amount a muscle can stretch, not the actual length of a muscle.

- ☐ **Short Muscle:** Muscle is not elongated enough – or is too tight – to provide adequate force production.
- ☐ **Long Muscle:** Muscle is too elongated – or stretched too long – to provide adequate force production. This is often termed an “inhibited” muscle.

If a muscle is shortened or lengthened too much, force production of the muscle will diminish to the point where minimal or no force production is possible (22). The relationship between a muscle's length and the ability to produce force resembles a bell curve.

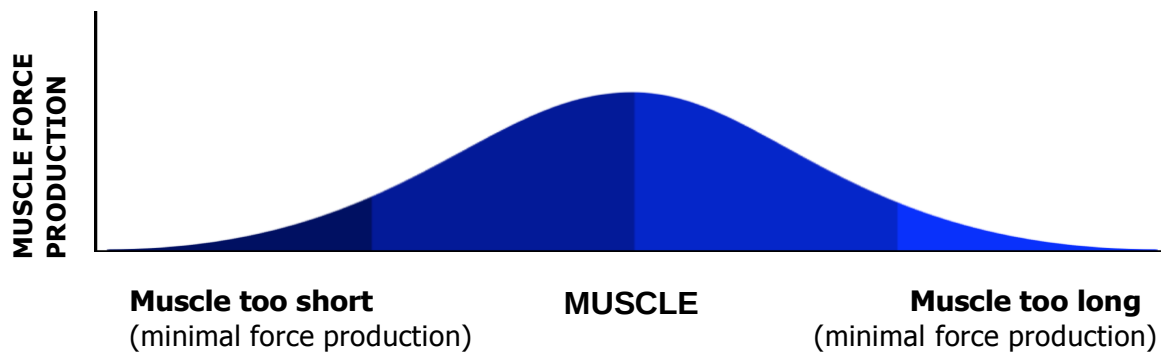


Figure 3.45 Muscle length/tension relationship

This is an important relationship to consider as it works hand in hand with biomechanics. No matter how great a training program may be, if a runner is not biomechanically efficient enough to maximize force production, the person will not perform optimally.

As an example of this and as noted earlier, let us look at bicycle seat height and hamstring activation. If the seat is too high, the hamstrings will become so elongated, minimal or no force production will be possible. Conversely, if the seat is too low the hamstrings will be hardly stretched at all, so force production from this muscle group will be negated almost completely.

Reduced Muscle Activation

When a muscle is too short or too long, impulses from the nervous system to the muscle are decreased. This decrease also affects the ability of the opposing (antagonist) muscle to contract properly. As you can see, this relates directly back to the muscle length/tension relationship (22).

For example, let us say the tibialis anterior (shin) muscle of a new runner is extremely tight (i.e., short) because of an inefficient running gait. As the tibialis anterior is responsible for dorsiflexion of the foot, this tightness will also inhibit optimal contraction of the soleus and gastrocnemius (calf muscles) muscles, as

they are the antagonists to the tibialis anterior and responsible for plantar flexion. This muscle contractile imbalance will not only affect the gait of the runner, but could possibly lead to injury if not addressed.

Fatigue is also a factor regarding reduced muscle activation. Muscle unit recruitment and firing rate decreases with fatigue (562, 563).

Additionally, muscle coactivation decreases the muscular force of agonist and antagonist muscles that surround a joint (564). It is theorized that a benefit of muscle coactivation is to stabilize a joint, as the opposing muscle forces of the antagonist and agonists effectively cancel each other out (565).

Muscular Compensation

This goes hand in hand with *reduced muscle activation*. Muscular compensation is typically caused by reduced muscle activation and results from a muscle (e.g., prime mover muscle) not being able to produce the required force. Therefore, a synergist or support muscle takes over the primary muscle's responsibility.

As noted earlier, for many endurance athletes, muscle activation of the gluteal muscles is reduced and the hamstrings assume force production. This leads to premature fatigue of the hamstrings as well as potential overuse injuries of the hamstrings as they are supposed to be synergist muscles to the glutes, not the primary movers. Nobody likes to do two jobs while getting paid for only one – muscles are no different!

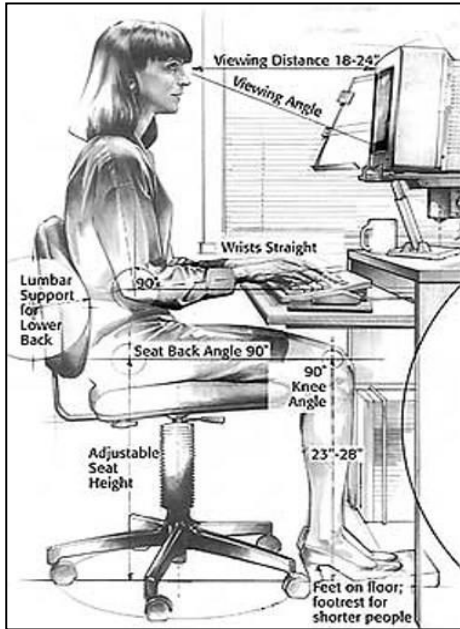


Figure 3.46 Correct ergonomics

Computer Workstation Variables
Integrated Safety Management, Berkeley Lab. July
2008

Workplace Postural Issues

The type of job that your client has is often reflected in what areas of the body are imbalanced. Many of your clients likely have desk jobs. This means for eight or more hours a day, they are in a seated position. If you look at the typical seated position, you will often see the following postural issues:

Hamstrings: shortened

Hip Flexors: shortened

Spine: excessive spinal flexion in the lumbar and thoracic regions, which often results in excessive cervical spinal extension

Shoulders: internally rotated and elevated

Core: weak inner unit core musculature

As you can imagine, no matter what you do during your one-on-one sessions with your client, unless you have them stretch/strength train on their own and change their workplace posture, these aforementioned issues will not improve. This is why it is so important for your client to have an ergonomically correct workstation. *Ergonomics* is the science of adapting postural and equipment changes at a workstation. For companies, the result of employees with poor workstation configurations is often decreased productivity due to lower back, neck, and wrist/forearm pain. For these reasons, many companies employ certified ergonomic specialists whose role is to ensure the workforce is set up at workstations properly.

❖❖ SUMMARY

- ☐ Muscles do not have their own “memory.”
- ☐ A concentric muscle contraction relates to the shortening of a muscle.
- ☐ An eccentric muscle contraction relates to the lengthening of a muscle.
- ☐ Hypertrophy is an increase in muscle size.
- ☐ Atrophy is a decrease in muscle size.
- ☐ There are two types of muscle contractions: *voluntary* and *involuntary*.
- ☐ Up to 70 percent of body heat is generated by the muscles.
- ☐ The structure of skeletal muscle can be viewed as a thick cable that is made up of many smaller cables.
- ☐ *Actin* and *myosin* allow the sarcomere to contract.
- ☐ The *sliding filament model* demonstrates how sarcomeres and thus how a muscle contracts.
- ☐ Understanding a muscle’s origin and insertion is important in determining the muscle’s effect on the body.
- ☐ Type I muscle fibers are oxidative and are the muscle fiber primarily used by runners.
- ☐ Hypertrophy refers to the increase in size of a muscle fiber whereas hyperplasia refers to the increase in the number of muscle fibers.
- ☐ Nerve innervation (electric stimulation) of muscles is what makes a muscle contract.
- ☐ A muscle fiber cannot partially contract.
- ☐ Tendons connect muscle to bone.
- ☐ Ligaments connect bone to bone.
- ☐ Fascia permeates the entire body, and based on the tension of the fascia, it can cause pain and postural abnormalities.
- ☐ Core musculature is divided into an *inner unit* and *outer unit*.
- ☐ *Muscle burn* is the result of a buildup of acidity in the muscle cells. More specifically, this acidity is caused by the release of hydrogen ions during the fast turnover of ATP.
- ☐ DOMS largely results from eccentric muscle contractions.
- ☐ There should be no more than 15 minutes from the end of a warm-up period to the beginning of a running session.
- ☐ As a coach, you are not allowed to advise or prescribe any medications to your client – including NSAIDs.
- ☐ Muscles work together in a set pattern to elicit body movement in what is called the *kinetic chain*.

- Muscle imbalances are characterized by a lack of balance between two muscles that surround a joint.
- Exercise-induced muscle cramps are the result of muscle fatigue, not dehydration or low electrolyte levels.
- If a muscle is too long or too short, the muscle's force production will be greatly diminished.
 - If a muscle's force production is greatly diminished, this often requires another muscle(s) to assist in force production. This is termed *muscular compensation*.
- Poor workplace posture often creates and magnifies muscle imbalances.